Autonomous Driving Safety Analyst

Technical RAG assistant for ISO 26262, ISO 21448 (SOTIF), ISO 8800, NCAP/IIHS, Hazard and Risk Analysis (HARA), AI safety evidence, scenario analysis, item safety cases, and video-grounded evidence review.

OpenAI premium path

Local Qwen baseline

LoRA fine-tuning experiment

Standards + video RAG

Whisper STT + optional TTS

Evaluation

Harn Zhin Chong

HOOK + OVERVIEW

Why this is more than a chatbot

Problem

• Creating and reviewing safety-case work products across multiple standards is time-consuming, repetitive, and easy to make inconsistent across functions.

- Overlapped safety standards (core behavior safety standards):
  ISO 26262,
  ISO 21448 (SOTIF),
  ISO 8800

Solution

• A Streamlit safety analyst grounded in project evidence.

• RAG over standards, examples, HARA profiles, dataset notes, and video transcripts.

• Three model paths: OpenAI (GPT-4o), local Qwen, and Qwen LoRA experiment.

Portfolio value

• RAG architecture and prompt conditioning.

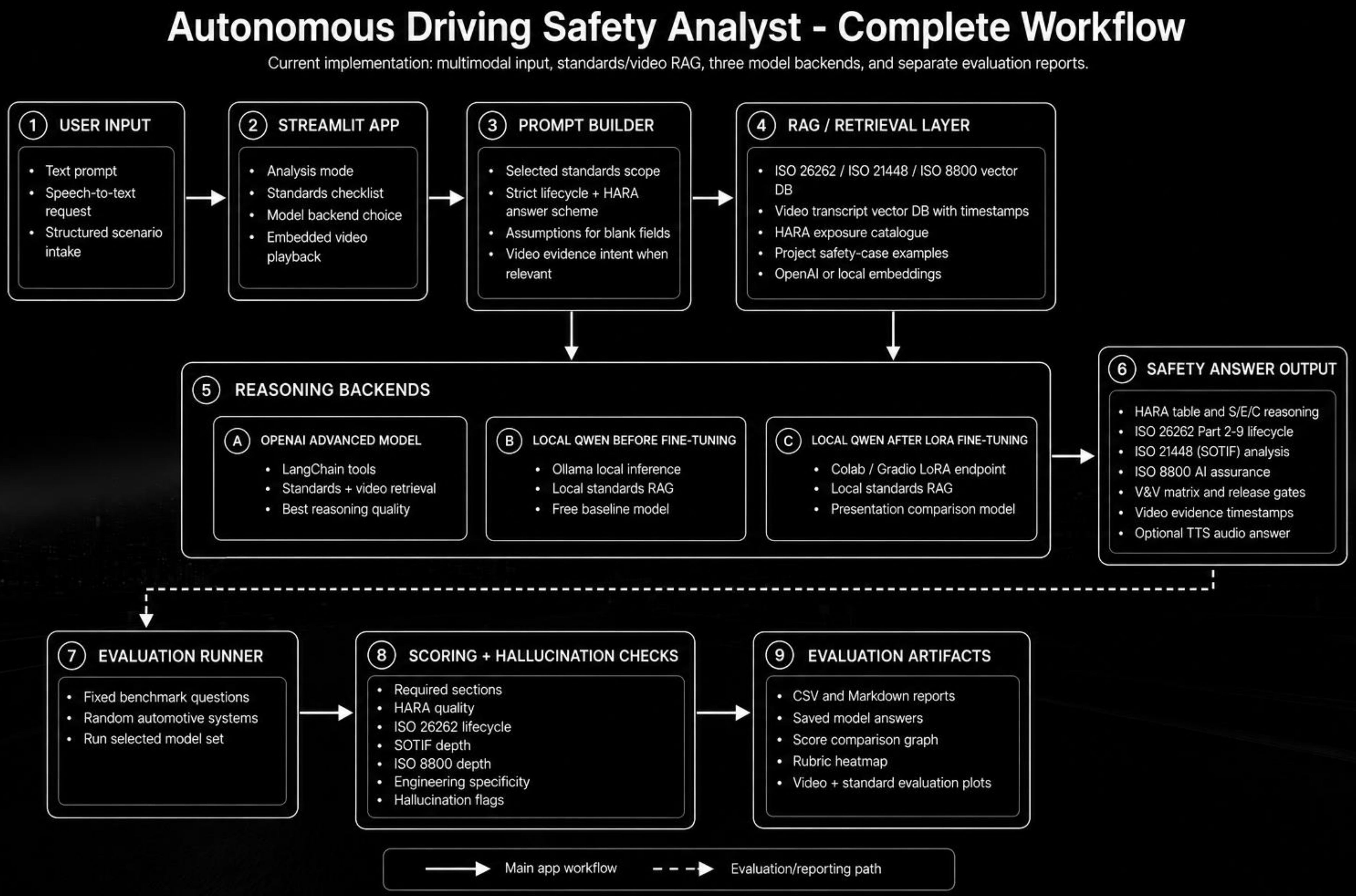
• Local/private model integration.

• Fine-tuning workflow in Colab.

• Evaluation with rubric, plots, saved answers, and hallucination checks.

TECHNICAL SUMMARY

End-to-end project workflow



DATASET

Sources, preprocessing, chunking, and vector storage

Source	Processing	Storage / Use
ISO Standards (ISO 26262. ISO 21448 (SOTIF) and ISO 8800) + Synthetic evaluation schemes	Markdown/PDF parsing, chunking, metadata tagging	Chroma standards DB for ISO 26262, ISO 21448, ISO 8800
Safety-case examples (Synthetic safety-case examples)	Curated worked examples for AEB, lane keeping, LiDAR perception	Retrieved as project-specific evidence
HARA + dataset profiles (nuScene dataset from Kaggle)	Structured risk/evidence snippets	Used for S/E/C rationale and ISO 8800 data risks
YouTube video transcripts (84 videos)	Transcript ingestion with timestamps	Video DB for evidence snippets and playback timestamps

```
python ingest_all.py
# builds standards DB and video transcript DB


Chroma metadata: source, standard, section, video_id, timestamp
```

- Paid path can combine standards + video evidence.
- Local path can use local standards embeddings to reduce embedding cost.
- Each answer states assumptions and evidence limits.

Q&A AGENT

Prompt builder and answer contract

Prompt conditioning

- Scopes the answer to selected standards.
- Injects a strict lifecycle and HARA answer scheme.
- Handles blank structured fields by stating assumptions.
- Adds video-evidence intent when the question asks for evidence or failure analysis.

```
LOCAL_STRICT_ANSWER_SCHEME = """
1. Opening Map
2. Item Definition
3. Functional Decomposition
4. HARA Screening
...
12. Final Safety Argument
"""
```

Why this matters

- Safety answers must be complete, not conversational summaries.
- The schema makes model comparison possible.
- The evaluator can check required sections and standards coverage.

TOOLS

RAG, audio, video, local model serving, and evaluation

Tool / component	Purpose
Chroma vector DB	Stores standards chunks, examples, HARA evidence, and video transcript chunks
OpenAI tools agent	Premium reasoning path with tool calls and review pass
Ollama	Runs local Qwen baseline before fine-tuning
Google Colab + Gradio	Serves the Qwen LoRA adapter endpoint
Whisper STT	Speech request input through uploaded/recorded audio
OpenAI TTS	Optional audio answer output
Evaluation scripts	Generate reports, saved answers, score plots, and rubric heatmaps

REASONING BACKENDS

Why the app supports three model paths

Backend	Strength	Trade-off	Best use
OpenAI	Best reasoning, tools, video + standards retrieval	Paid API cost	Final/premium analysis
Local Qwen base	Free/private path, strong with RAG + answer schema	Slower; no hosted API by default	Cost-sensitive use
Qwen LoRA	Shows fine-tuning workflow and adapter learning	Colab endpoint/Out of Memory (OOM) risk; small(30) synthetic data	Portfolio experiment

Key lesson

- Fine-tuning is not automatically better than RAG.
- Local Qwen base performs strongly when prompt + retrieval + guardrails are disciplined.

```
if model_key == "openai": run_agent_tools()
if model_key == "local_base": run_ollama_rag()
if model_key == "local_lora": call_gradio_lora()
```

EVALUATION

Transparent scoring for standards answers

How it runs

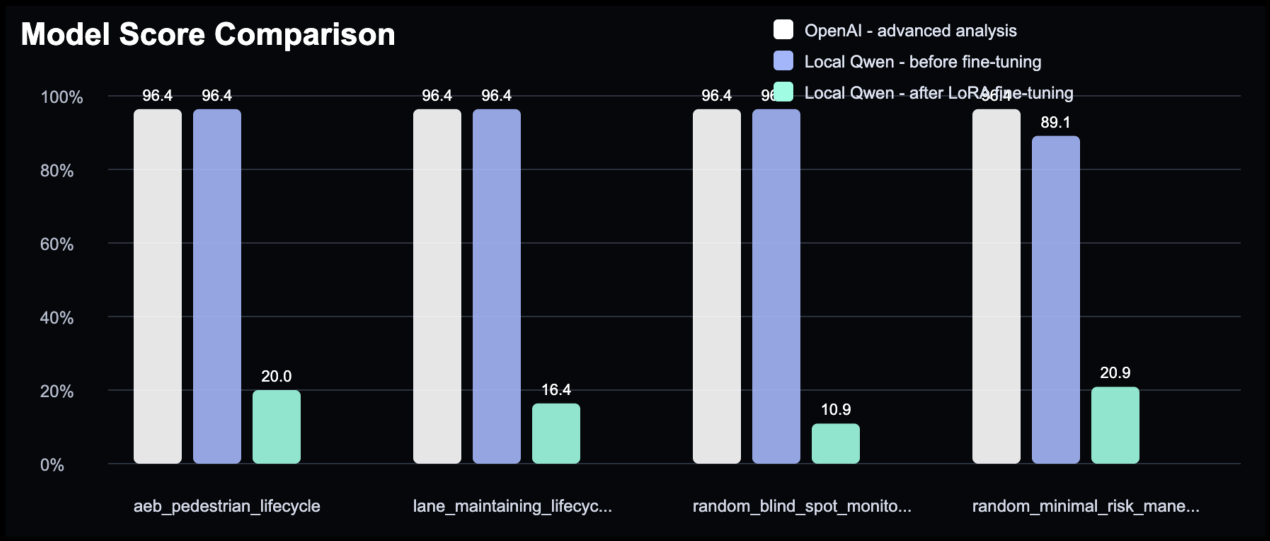
- Runs fixed benchmark questions from evaluation/test_questions.jsonl.
- Adds random automotive systems when requested.
- Saves every generated answer for manual inspection.
- Scores with deterministic 110-point rubric normalized to percent.
- Flags unsupported clause-like claims and missing evidence limits.

```
.venv/bin/python evaluation/evaluate_models.py \
  --models openai local_base local_lora \
  --limit 2 --random-cases 2 --seed 42
```

Rubric category	Max
Required sections	20
HARA quality	20
ISO 26262 lifecycle	20
SOTIF depth	15
ISO 8800 depth	15
Engineering specificity	10
Hallucination control	10

MODEL EVALUATION RESULTS

Latest standards-only benchmark



Rubric Category Heatmap

	Sections	HARA	ISO 26262	SOTIF	ISO 8800	Engineering	Hallucination
aeb_pedestrian_lifecycle | openai	20/20	16/20	20/20	15/15	15/15	10/10	10/10
aeb_pedestrian_lifecycle | local_b...	20/20	16/20	20/20	15/15	15/15	10/10	10/10
aeb_pedestrian_lifecycle | local_l...	2/20	0/20	0/20	12/15	0/15	8/10	0/10
lane_maintaining_lifecycle | opena...	20/20	16/20	20/20	15/15	15/15	10/10	10/10
lane_maintaining_lifecycle | local...	20/20	16/20	20/20	15/15	15/15	10/10	10/10
lane_maintaining_lifecycle | local...	0/20	0/20	0/20	0/15	0/15	8/10	10/10
random_blind_spot_monitoring_syste...	20/20	16/20	20/20	15/15	15/15	10/10	10/10
random_blind_spot_monitoring_syste...	20/20	16/20	20/20	15/15	15/15	10/10	10/10
random_blind_spot_monitoring_syste...	0/20	4/20	0/20	0/15	0/15	8/10	0/10
random_minimal_risk_maneuver_contr...	20/20	16/20	20/20	15/15	15/15	10/10	10/10
random_minimal_risk_maneuver_contr...	18/20	16/20	20/20	15/15	15/15	10/10	4/10
random_minimal_risk_maneuver_contr...	0/20	0/20	5/20	0/15	0/15	8/10	10/10

Average evaluation result for each model

OpenAI **96.4%**

Local Qwen base **94.6%**

Qwen LoRA **17.1%**

Artifacts: model_eval_20260528_094923.csv / .md, saved answers, score plot, rubric heatmap.

Interpretation

- OpenAI and local Qwen base both score strongly on the strict answer schema.
- LoRA results are kept honest: current endpoint/answer quality is not strong enough for final use.
- This is useful evidence that RAG + prompt discipline can outperform a small fine-tune.

PAID FEATURE EVALUATION

Video evidence + standards reasoning

Video + Standards Rubric Heatmap

	Video Evidence	Video Grounding	Standards	Classification	Engineering	Limits	Hallucination
video_failure_plus_standards	20/20	15/15	20/20	15/15	15/15	5/10	5/5
video_pedestrian_edge_case	20/20	15/15	20/20	15/15	15/15	5/10	5/5
video_weather_visibility_cas...	20/20	15/15	20/20	15/15	15/15	5/10	5/5

Why it matters

- This is the premium selling point: standards + real video transcript evidence.
- Answers include video timestamps, safety interpretation, engineering actions, and evidence limitations.
- Latest sample score: 95.0% average across three cases.

```
.venv/bin/python evaluation/evaluate_video_standard.py
```

Outputs: video_standard_eval_20260528_110341.csv/.md, saved answers, video-standard score graph, heatmap.

Model Comparison Summary

Path	How it reasons	Strength	Current limitation
OpenAI	LangChain tool-calling agent over standards + video tools	Best quality, dynamic retrieval, paid feature path	Uses paid API and external service
Local Qwen Base	Pre-retrieved local standards context + strict prompt	Free/private path, strong with RAG guardrails	No direct tool loop, slower local generation
Qwen LoRA	Colab-served adapter + local RAG prompt context	Proves fine-tuning workflow	Only 30 synthetic examples; endpoint/GPU instability

Takeaway

- The strongest product story is OpenAI for premium standards+video evidence, local Qwen base for free/private standards reasoning, and LoRA as a future fine-tuning improvement path.

CONCLUSION

Closing argument and demo script

Final message

- This is a standards-aware safety analysis workflow, not a generic chatbot.
- It demonstrates RAG, tool use, local model integration, LoRA experimentation, audio input/output, video evidence, and quantitative evaluation.
- The strongest engineering lesson: structured retrieval and evaluation matter as much as model choice.

Architecture Annex

End-to-End Architecture: User Request to Output

Live app path

1 User Request
- text prompt
- speech input
- scenario form

2 Streamlit App
- mode selector
- standards selector
- model selector

3 Prompt Builder
- scope standards
- strict answer scheme
- assumptions

4 RAG Context
- standards DB
- video transcript DB
- examples/profiles

5 Reasoning Backend
- OpenAI tools
- local Qwen
- LoRA endpoint

6 Structured Output
- safety case
- tables
- evidence

Evaluation is a separate reproducibility path, not part of every live answer

7 Evaluation Runner
- fixed benchmark cases
- random systems
- selected model list

8 Rubric + Checks
- required sections
- HARA quality
- ISO/SOTIF/ISO 8800 depth
- hallucination flags

9 Artifacts
- CSV report
- Markdown report
- saved answers
- graphs

Shared Front-End + Prompt Conditioning

1 Input Channels
- typed request
- speech-to-text request
- structured scenario form

2 Streamlit State
- chat history
- selected standards
- backend selection
- voice output toggle

3 Scoped Prompt
- adds ISO scope
- preserves original question
- allows unknown fields

4 Answer Contract
- opening map
- functional decomposition
- HARA S/E/C
- lifecycle + V&V
- final safety argument

Why this matters
- prevents short chatbot summaries
- makes missing assumptions explicit
- creates a common answer shape for all models

Backend-neutral design
- OpenAI, Qwen base, and LoRA receive the same task intent
- model differences become easier to evaluate

Output hooks
- optional TTS
- video evidence cards
- timestamp playback

1st Model:
OpenAI Path: LangChain Tool-Calling Agent

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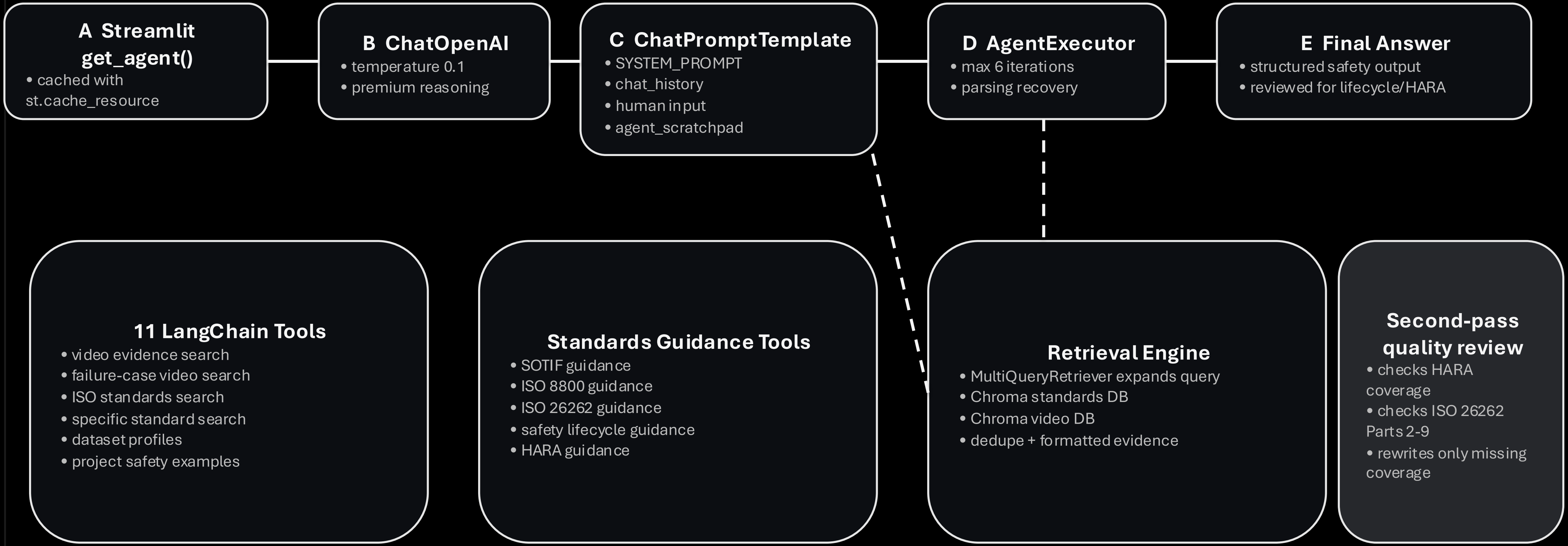

2nd Model:
Local Qwen Base Path: Pre-Retrieval RAG

Local Qwen Base Path: Pre-Retrieval RAG

1 User Question
- same scoped prompt

2 Local Retrieval Planner
- selects standards queries
- no tool-calling loop

3 Local Context Pack
- ISO evidence
- SOTIF / ISO 8800 guidance
- project examples

4 Ollama Qwen
- local inference
- free/private path

5 Completeness Guard
- same-model revision if sections missing

Why not direct LangChain tools?
- local model tool calls can be malformed
- pre-retrieval is more stable for demos
- keeps context deterministic

What it still uses
- local standards Chroma DB
- local embeddings for free model
- same answer schema as OpenAI

Result
- strong baseline performance
- private/local option
- no video DB in local answer path

3rd Model:
Fine-Tuned LoRA Path: Experiment and Serving Flow

Fine-Tuned LoRA Path: Experiment and Serving Flow

1 Synthetic SFT Data
- 30 teacher-generated examples
- from seed prompts + retrieved evidence

2 Colab Training
- Qwen LoRA adapter
- GPU training

3 Adapter Folder
- qwen_safety_lora
- saved to Drive/local

4 Gradio Endpoint
- temporary public URL
- LOCAL_LORA_API_URL

5 Streamlit Call
- gradio_client
- returns answer

Runtime safeguards
- invalid numeric/debug output detection
- large output token setting
- error message if endpoint unreachable

Experiment result
- pipeline works
- small synthetic data is not enough
- not production-quality yet

Presentation position
- show as fine-tuning experiment
- local Qwen base remains stronger free path
- future work: curated data + stable GPU

Evaluation Architecture

Evaluation Architecture: Standards + Video Evidence

1 Standards Evaluation
- evaluate_models.py
- OpenAI + local base + LoRA
- fixed + random systems

2 Video + Standards Evaluation
- evaluate_video_standard.py
- OpenAI paid path
- video transcript + ISO reasoning

3 Answer Generation
- same app model functions
- saved Markdown answer per case

4 Rubric Scoring
- required sections
- HARA / lifecycle depth
- video evidence use
- hallucination control

5 Reports + Graphs
- CSV
- Markdown summary
- score bar charts
- rubric heatmaps

Evaluation principle
- The evaluation path is separate from the live chat path so model quality can be compared with saved answers, reproducible prompts, and visible scoring criteria.